



Center and the Effect of an Outlier

Grades 9-12 • SAT / ACT mathematics foundations

Problem-Solving and Data Analysis • Compare how one changed value affects center and spread.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Compare how one changed value affects center and spread.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. Data: [4, 6, 8, 10, 12]. Replace the greatest value with 27. How do the mean, median, and range change?

Only the total changes by 15, so mean increases by $15/5 = 3$. The third ordered value remains 8. The maximum, and therefore range, increases by 15.

Model 2. Data: [9, 11, 13, 15, 17]. Replace the greatest value with 57. How do the mean, median, and range change?

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 13. The maximum, and therefore range, increases by 40.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 29. How do the mean, median, and range change?

2. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 44. How do the mean, median, and range change?

3. Data: [5, 7, 9, 11, 13]. Replace the greatest value with 23. How do the mean, median, and range change?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. Data: [2, 4, 6, 8, 10]. Replace the greatest value with 30. How do the mean, median, and range change?

5. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 49. How do the mean, median, and range change?

6. Data: [3, 5, 7, 9, 11]. Replace the greatest value with 31. How do the mean, median, and range change?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 54. How do the mean, median, and range change?

8. Data: [5, 7, 9, 11, 13]. Replace the greatest value with 48. How do the mean, median, and range change?

9. Data: [2, 4, 6, 8, 10]. Replace the greatest value with 30. How do the mean, median, and range change?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. Data: [4, 6, 8, 10, 12]. Replace the greatest value with 52. How do the mean, median, and range change?

11. Data: [4, 6, 8, 10, 12]. Replace the greatest value with 52. How do the mean, median, and range change?

12. Data: [2, 4, 6, 8, 10]. Replace the greatest value with 50. How do the mean, median, and range change?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. Data: [3, 5, 7, 9, 11]. Replace the greatest value with 31. How do the mean, median, and range change?

14. Data: [8, 10, 12, 14, 16]. Replace the greatest value with 56. How do the mean, median, and range change?

15. Data: [9, 11, 13, 15, 17]. Replace the greatest value with 62. How do the mean, median, and range change?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. Data: [7, 9, 11, 13, 15]. Replace the greatest value with 40. How do the mean, median, and range change?



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. Data: [4, 6, 8, 10, 12]. Replace the greatest value with 52. How do the mean, median, and range change?

2. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 24. How do the mean, median, and range change?

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. Data: [3, 5, 7, 9, 11]. Replace the greatest value with 31. How do the mean, median, and range change?

4. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 44. How do the mean, median, and range change?

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. Data: [5, 7, 9, 11, 13]. Replace the greatest value with 38. How do the mean, median, and range change?

6. Data: [2, 4, 6, 8, 10]. Replace the greatest value with 30. How do the mean, median, and range change?

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. Data: [3, 5, 7, 9, 11]. Replace the greatest value with 36. How do the mean, median, and range change?

2. Data: [3, 5, 7, 9, 11]. Replace the greatest value with 31. How do the mean, median, and range change?



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. Data: [9, 11, 13, 15, 17]. Replace the greatest value with 62. How do the mean, median, and range change?

4. Data: [6, 8, 10, 12, 14]. Replace the greatest value with 39. How do the mean, median, and range change?



Practice worked solutions

1. Mean +3; median unchanged; range +15

Only the total changes by 15, so mean increases by $15/5 = 3$. The third ordered value remains 10. The maximum, and therefore range, increases by 15.

2. Mean +6; median unchanged; range +30

Only the total changes by 30, so mean increases by $30/5 = 6$. The third ordered value remains 10. The maximum, and therefore range, increases by 30.

3. Mean +2; median unchanged; range +10

Only the total changes by 10, so mean increases by $10/5 = 2$. The third ordered value remains 9. The maximum, and therefore range, increases by 10.

4. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 6. The maximum, and therefore range, increases by 20.

Practice worked solutions

5. Mean +7; median unchanged; range +35

Only the total changes by 35, so mean increases by $35/5 = 7$. The third ordered value remains 10. The maximum, and therefore range, increases by 35.

6. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 7. The maximum, and therefore range, increases by 20.

7. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 10. The maximum, and therefore range, increases by 40.

8. Mean +7; median unchanged; range +35

Only the total changes by 35, so mean increases by $35/5 = 7$. The third ordered value remains 9. The maximum, and therefore range, increases by 35.

Practice worked solutions

9. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 6. The maximum, and therefore range, increases by 20.

10. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 8. The maximum, and therefore range, increases by 40.

11. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 8. The maximum, and therefore range, increases by 40.

12. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 6. The maximum, and therefore range, increases by 40.

Practice worked solutions

13. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 7. The maximum, and therefore range, increases by 20.

14. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 12. The maximum, and therefore range, increases by 40.

15. Mean +9; median unchanged; range +45

Only the total changes by 45, so mean increases by $45/5 = 9$. The third ordered value remains 13. The maximum, and therefore range, increases by 45.

16. Mean +5; median unchanged; range +25

Only the total changes by 25, so mean increases by $25/5 = 5$. The third ordered value remains 11. The maximum, and therefore range, increases by 25.

Transfer worked solutions

1. Mean +8; median unchanged; range +40

Only the total changes by 40, so mean increases by $40/5 = 8$. The third ordered value remains 8. The maximum, and therefore range, increases by 40.

2. Mean +2; median unchanged; range +10

Only the total changes by 10, so mean increases by $10/5 = 2$. The third ordered value remains 10. The maximum, and therefore range, increases by 10.

3. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 7. The maximum, and therefore range, increases by 20.

4. Mean +6; median unchanged; range +30

Only the total changes by 30, so mean increases by $30/5 = 6$. The third ordered value remains 10. The maximum, and therefore range, increases by 30.

Transfer worked solutions

5. Mean +5; median unchanged; range +25

Only the total changes by 25, so mean increases by $25/5 = 5$. The third ordered value remains 9. The maximum, and therefore range, increases by 25.

6. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 6. The maximum, and therefore range, increases by 20.

Exit-ticket worked solutions

1. Mean +5; median unchanged; range +25

Only the total changes by 25, so mean increases by $25/5 = 5$. The third ordered value remains 7. The maximum, and therefore range, increases by 25.

2. Mean +4; median unchanged; range +20

Only the total changes by 20, so mean increases by $20/5 = 4$. The third ordered value remains 7. The maximum, and therefore range, increases by 20.

3. Mean +9; median unchanged; range +45

Only the total changes by 45, so mean increases by $45/5 = 9$. The third ordered value remains 13. The maximum, and therefore range, increases by 45.

4. Mean +5; median unchanged; range +25

Only the total changes by 25, so mean increases by $25/5 = 5$. The third ordered value remains 10. The maximum, and therefore range, increases by 25.



Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Compare how one changed value affects center and spread. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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